

Department of Chemical Engineering
Chemical Reaction Engineering II (CL24303)

MO2026
Tutorial 2

Assigned on	20.08.2026	Submission Date and Time	30.08.2026 23:59
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1. H₂ and N₂ are known to undergo dissociative chemisorption on the Fe catalyst in the Haber-Bosch process. Show that at equilibrium: 2

$$C_{N,S} = \frac{C_t \sqrt{K_{N_2} p_{N_2}}}{1 + 2\sqrt{K_{N_2} p_{N_2}} + 2\sqrt{K_{H_2} p_{H_2}}}$$

2. The following data on the adsorption of CO₂ on activated carbon and silica at 25 °C were reported in JACS (1950) 72 (3): 1153–1157. Assuming the Langmuir Adsorption isotherm (Eq. 1) to be valid, use non-linear regression in MS Excel to determine the maximum adsorbate loading - q_{max} and the adsorption equilibrium constant K .

$$\theta = \frac{q}{q_{max}} = \frac{Kp}{1 + KP} \quad (1)$$

Activated carbon		Silica	
p(mm)	q (mmol/gm)	p(mm)	q (mmol/gm)
15.7	0.791	11.1	0.0564
44.0	1.141	25	0.1252
54.2	1.223	43.5	0.1980
84	1.416	71.4	0.2986
107	1.532	100	0.3850
139	1.646	158.9	0.5441
177	1.769	227.5	0.7020
215.5	1.872	304.2	0.843
278	2.006	387	1.010
331.5	2.105	468	1.138
429.5	2.254	569	1.288
536	2.373	677.8	1.434
585.7	2.454	775	1.562
640.7	2.479		
680.2	2.510		

3. Adsorption data for CO₂ and CH₄ on zeolite -13X at 298,308, and 323K are provided in Tables 2 (*J. Chem. Eng. Data* 2004, 49, 1095-1101). Using non-linear regression in MS Excel, fit the equilibrium data to the Toth isotherm (Eq. 1) and the multi-site Langmuir (Eq. 2) isotherm and determine, $q_{i,max}$, K_i , n_i , and a_i . Thereafter, determine the ΔH_{ads} by fitting a linear equation to the van't Hoff equation

$$\theta = \frac{q_i}{q_{i,max}} = \frac{K_i p_i}{[1 + (K_i p_i)^{n_i}]^{1/n_i}} \quad (2)$$

$$\theta = \frac{q_i}{q_{i,max}} = K_i p_i \left(1 - \frac{q_i}{q_{i,max}}\right)^{a_i} \quad (3)$$

Table 1 Experimental Data of CH ₄ Adsorption Equilibrium on Zeolite 13X					
T = 298 K		T=308 K		T=323 K	
P [MPa]	q [mol/kg]	P [MPa]	q [mol/kg]	P [MPa]	q [mol/kg]
0.00000	0.000	0.00000	0.000	0.00000	0.000
0.00405	0.024	0.00525	0.022	0.00603	0.017
0.01203	0.089	0.01115	0.064	0.01215	0.052
0.01910	0.131	0.02040	0.109	0.02200	0.090
0.05515	0.326	0.04007	0.210	0.03302	0.140
0.125	0.712	0.08516	0.415	0.04612	0.196
0.165	0.877	0.135	0.623	0.05505	0.227
0.210	1.120	0.135	0.632	0.08010	0.312
0.306	1.474	0.190	0.823	0.115	0.432
0.345	1.617	0.280	1.133	0.165	0.590
0.425	1.830	0.310	1.232	0.240	0.731
0.631	2.357	0.350	1.360	0.340	1.009
0.819	2.726	0.445	1.618	0.340	1.009
1.070	3.060	0.585	1.931	0.400	1.193
1.180	3.260	0.585	1.932	0.510	1.423
1.410	3.530	0.695	2.154	0.505	1.395
1.720	3.834	0.780	2.342	0.635	1.653
1.890	3.991	0.875	2.466	0.780	1.929
2.175	4.198	1.110	2.792	0.865	2.077
2.610	4.506	1.480	3.201	0.955	2.211
2.985	4.750	1.700	3.409	1.185	2.545
3.365	4.987	1.790	3.480	1.495	2.890
3.560	5.103	2.170	3.781	1.695	3.067
3.745	5.191	2.530	4.038	1.860	3.169
3.745	5.191	2.535	4.034	2.425	3.577
4.260	5.469	2.840	4.236	2.570	3.670
4.725	5.719	3.640	4.702	3.015	3.933
		4.015	4.884	3.425	4.199
		4.490	5.127	3.425	4.200
		4.720	5.234	3.670	4.341
				4.180	4.585
				4.445	4.706
				4.745	4.830

Table 2 Experimental Data of CO ₂ Adsorption Equilibrium on Zeolite 13X					
T = 298 K		T=308 K		T=323 K	
P [MPa]	q [mol/kg]	P [MPa]	q [mol/kg]	P [MPa]	q [mol/kg]
0.00000	0.000	0.00000	0.000	0.00000	0.000
0.00118	1.147	0.00126	0.825	0.00106	0.356
0.00610	2.249	0.00407	1.458	0.00214	0.529
0.02905	3.659	0.00905	2.060	0.00501	1.060
0.08610	4.500	0.01325	2.400	0.00907	1.430
0.160	5.060	0.02007	2.734	0.02423	2.090
0.310	5.580	0.04515	3.380	0.04220	2.490
0.525	6.040	0.09503	4.050	0.07503	2.900
1.015	6.520	0.160	4.490	0.145	3.400
1.015	6.500	0.220	4.740	0.270	3.915
1.445	6.920	0.280	4.960	0.390	4.100
1.935	6.960	0.370	5.167	0.545	4.329
2.280	7.090	0.485	5.315	0.705	4.532
2.660	7.220	0.570	5.399	0.850	4.740
3.200	7.372	0.700	5.586	1.010	4.820
		0.875	5.718	1.175	4.930
		0.875	5.718	1.455	5.200
		1.015	5.803	1.720	5.240
		1.200	5.942	2.125	5.430
		1.505	6.116	2.695	5.620
		1.830	6.291	3.395	5.762
		2.235	6.510		
		2.500	6.631		
		2.860	6.769		
		3.065	6.885		
		3.065	6.820		
		3.365	6.920		